Within-sample cell-type deconvolution (group-agnostic)

Per sample × per cell type: is H3K27ac cfDNA enriched at this cell type's signature, above its own local genomic background?

Fragment BED (1 / sample)
midpoint = (start+end)/2

RRE universe (334k regions)
cell-type–exclusive H3K27ac signatures

chromHMM off-target BED (220k)
never-active windows → null territory

Signature regions per cell type
exclusive to one type in the panel

Background tiles
subtract signatures → 5 kb windows

QC / null checks
permuted-signature null
+ anti-signal (expect log2FE ≈ 0)

Count fragment midpoints per region (bedtools intersect, parallel over samples)
compute GC content per region (bedtools nuc)

Sadeh local background rate
trimmed-mean density, genome → chr → 2 Mb hierarchy

GC × length matched background
10 tiles / signature region drawn from matching bin

+ per-sample NB dispersion α (from background variance)

Negative-binomial GLM (per sample × cell type)
counts ~ 1 + is_signature offset = log(local expected count)
β_1 = log-enrichment of signature over its local background → one-sided upper-tail p

BH-FDR within sample
enriched = q < 0.05

log2 fold-enrichment
pooled obs / expected counts

Fractional contribution
excess share across cell types

Input data · Region preparation · Statistical model · Per-sample call / score · QC